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Theory-driven design of high-valence metal sites for water oxidation confirmed using *in situ* soft X-ray absorption

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The efficiency with which renewable fuels and feedstocks are synthesized from electrical sources is limited at present by the sluggish oxygen evolution reaction (OER) in pH-neutral media. We took the view that generating transition metal sites with high valence at low applied bias should improve the activity of neutral OER catalysts. Using density functional theory, we find that the formation energy of desired Ni^{4+} sites is systematically modulated by incorporating judicious combinations of Co, Fe and non-metal phosphorus. Here we synthesized NiCoFeP oxyhydroxides and probed their oxidation kinetics by employing *in situ* soft X-ray absorption (sXAS). *In situ* sXAS studies of neutral-pH OER catalysts indicate ready promotion of Ni^{4+} under low overpotential conditions. NiCoFeP catalyst outperforms IrO_2 and retains its performance following 100 hours of operation. We showcase NiCoFeP in a membrane-free CO_2 electroreduction system that achieves a 1.99 V cell voltage at 10 mA/cm^2 , reducing CO_2 into CO and oxidizing H_2O to O_2 with a 64% electricity-to-chemical-fuel efficiency.

Electrochemical reduction of carbon dioxide to value-added fuels seeks to reduce the present-day dependence on legacy fossil fuels and thereby mitigate net CO_2 emissions¹⁻⁴. In electrochemical CO_2 reduction, the use of neutral-pH aqueous electrolyte is desired since it provides the highest CO_2 solubility while suppressing the undesired competing hydrogen evolution reaction.

Unfortunately, in pH-neutral electrolytes, the overpotential of the anodic oxygen evolution reaction (OER) is, in the best catalysts reported to date, very high ($> 460 \text{ mV}$ at 10 mA/cm^2)^{3, 5-7}. This limits the electrical-to-chemical power conversion efficiency of the overall reaction. To improve the energy efficiency of CO_2 conversion, efforts have

been made to provide CO₂ reduction in neutral media while operating OER in alkaline electrolyte. However, the effective overpotential remains high (~ 430 mV) due to the additional energy needed to overcome the effect of pH gradients^{3, 8, 9}, i.e. the need also to overcome the membrane potential.

Breakthroughs in low-overpotential, stable, and earth-abundant OER catalysts that operate optimally in neutral pH are therefore urgently needed to complement rapid advancements in CO₂ reduction electrocatalysts.

High-valence transition metal (TM) ions incorporated into metal oxides are known to achieve a high reactivity towards water oxidation^{5,10-15}. However, as evidenced in their Pourbaix diagrams, TM ions prefer to take on low-valence oxidation states at neutral pH^{5,11,16-18}.

We adopted a new strategy, one in which we would seek to minimize the energy (formation energy) required for active metal sites to reach the desired high oxidation state in neutral pH^{5,12}. We hypothesized that low active site formation energy would correlate well with a low overpotential to reach the high-turnover-frequency activated state.

We focus in on the well-known Ni²⁺ → Ni⁴⁺ earth-abundant catalyst^{5,12,18}. We sought to challenge our posited correlation experimentally, pursuing *in situ* monitoring of the oxidation state of the OER catalyst.

Multi-metal oxides based on Co, Fe, and Ni often outperform unary metal oxides due to modulation of their electronic properties and the resulting fine-tuning of the adsorption energies of the reaction intermediates^{10,19-25}. Concomitantly, cobalt-

phosphate compounds exhibit high activity at neutral pH^{6,11,26}. We reasoned that phosphate could help promote the transition metal to high valence states; and thus that the catalytic activity of promising multi-metal oxides (i.e. 3d metals) could be further enhanced if their 3d electronic structures were systematically and rationally modulated via incorporating multi-metal and non-metal modulators^{19,25,26}.

Here we therefore explored combinations of elements incorporated into Ni oxides in an atomically intimate manner¹⁹, seeking the ready formation of Ni⁴⁺ with the goal of improving the catalytic activity of the material. We synthesized NiCoFeP oxyhydroxides and probed their oxidation kinetics employing in situ soft X-ray absorption (sXAS). The new NiCoFeP catalysts exhibit exceptional OER performance, with a 330 mV overpotential at 10 mA/cm² in CO₂-saturated 0.5 M KHCO₃ electrolyte. The new catalyst outperforms the leading precious metal oxide IrO₂ and retains its performance following 100 hours of operation. We then combined the optimal multi-metal oxide with a gold nanoneedle CO₂ reduction catalyst to assemble an electrolyzer system which converts CO₂ and H₂O into CO and O₂. We found that the new system operated at a low cell voltage of 1.99 V and record electricity-to-chemical-fuel efficiency of 64%.

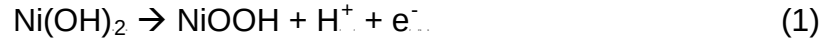
Results and Discussion

Theoretical studies. We began with computational studies aimed at understanding the effect of intimate metal and phosphorous doping on the stability of nickel structures with different valences. Density functional theory calculations with on-site Hubbard-like corrections (DFT+U) were performed on model Ni(OH)₂, β-NiOOH, and NiO₂ structures (doped with Co, Fe, and P) to model Ni²⁺, Ni³⁺, and Ni⁴⁺ oxidation states, respectively (see Supplementary Table 1 for optimized lattice constants). Experimentally, the NiO₂

phase is unstable and does not exist in the Ni Pourbaix diagram; simulations reproduce this observation, showing that NiO₂ formation is unfavorable by 0.31 eV/formula unit (fu).

However, once NiO₂ is doped with phosphorus or other metals, the formation energies become negative, indicating a more stable phase (Supplementary Table 2). Thus, these simulations suggest that doping using Co, Fe, and P can lower the Gibbs formation energy of the compounds containing Ni⁴⁺ that are otherwise unstable.

The oxidation of Ni²⁺ to Ni⁴⁺ is divided into the following proton-coupled electron transfer reactions²⁷:



We found that substitutional doping of nickel to phosphorus consistently lowered the Ni²⁺ → Ni³⁺ reaction free energies; while substitution of Ni with Co and Fe lowered the Ni³⁺ → Ni⁴⁺ reaction free energies (Fig. 1). Phosphorus, which prefers four-fold coordination, affects nearby hydrogen geometries and induces steric strain into the lattice, resulting in a destabilization of the Ni(OH)₂ phase. Moreover, Co and Fe in ternary blends modulate the electronic structure of the nickel oxides, stabilizing the Ni⁴⁺ phase.

In order to investigate the modulation of electronic structure with respect to transition metal doping, we carried out a density of states (DOS) analysis. It was found that the valence band maximum (VBM) was only 0.17 eV below the Fermi level for NiO₂ whereas the VBM was 0.20 eV below the Fermi level for Ni(OH)₂. This suggests that the Ni⁴⁺ shifts the Fermi level down compared to Ni²⁺ to expose hole states for

catalysis²⁸. From the DOS comparison between NiO_2 and $\text{Ni}_{1-x-y}\text{Co}_x\text{Fe}_y\text{O}_2$, we find that the NiCoFe oxide introduces more d-states within the gap compared to pure NiO_2 (Supplementary Fig. 1). The projected density of states (pDOS) analysis (Supplementary Fig. 2) reveals that electronic states near the Fermi level are predominantly from Ni and Fe 3d orbitals, suggesting that *both* Fe^{4+} and Ni^{4+} can act as catalytic active sites as they each contribute states near the valence band edge²⁹.

Oxidation-state changes monitored by *in situ* soft X-ray absorption. To challenge experimentally the concept that low formation energy could correlate with an early voltage onset of efficient OER, we sought a means to track the electronic states of metal sites *in situ*. Hard X-ray absorption (hXAS) has previously been used *in situ* to probe high-energy transitions between 1s and 4p electronic orbitals and is more relevant to local coordination environments. The electronically-relevant 3d final states is weakly evident in hXAS as a low-intensity pre-edge feature resulting from quadrupole transitions³⁰. For this reason, the hXAS pre-edges do not enable an informative determination of the narrow band, TM 3d electronic structure. Realtime tracking of the x-ray absorption near-edge region of the Ni K-edge was used to obtain initial information about their valence states and electronic configurations³¹.

We turned instead to soft X-ray absorption spectroscopy (sXAS) at TM $L_{2,3}$ -edges, which accurately and efficiently quantifies TM oxidation states since these dipole-allowed $2p \rightarrow 3d$ optical transitions probe directly the 3d-states that are crucial to chemical activities. It provides detailed information on the local electronic structure and is sensitive to chemical bonding and oxidation states³²⁻³⁷. The sXAS measurements performed here were made using partial fluorescence yield intensity. These spectra are

not strictly proportional to the x-ray attenuation coefficient due to state dependent decay probabilities³². These electron delocalization effects are known to enhance the fluorescence intensities of the higher energy transitions in a L-edge PFY spectrum. We considered these effects in our interpretation of *in situ* measurements. We concluded that the magnitude of the bias-dependent changes we observed was much greater than the small effects that arise due to state-dependent decay³³.

A challenge limiting the use of sXAS to study *in situ* operating OER catalysts arises from its limited probing depth. sXAS does not therefore reveal the (remote, far-side) active surface behavior of an operating catalyst when excited and probed through a window and substrate (Fig. 2).

We therefore synthesized highly nanoporous NiCoFeP samples into which we expected the electrolyte used in OER might substantially penetrate. We employed an *in situ* electrochemical flow cell that used an ultrathin Si₃N₄ membrane window (100 nm thick) to separate the liquid medium inside the cell from the high-vacuum environment needed for soft X-ray measurements (Fig. 2a). The working electrode for loading catalysts consisted of a ~20-nm Au film evaporated on the internal side of the Si₃N₄ membrane in contact with a flowing CO₂ saturated 0.5 M KHCO₃ solution as electrolyte. Scanning transmission electron microscopy (STEM) images of NiCoFeP reveal a structure composed of nanosheets and nanopores (Fig. 2b). Energy-dispersive X-ray spectroscopy (EDS) mapping (Supplementary Fig. 7) and electron energy loss spectroscopy (EELS) elemental maps with sub-nm resolution (Supplementary Fig. 9) show a uniform, uncorrelated spatial distribution of Ni, Co, Fe, and P with 9:1:1:0.1

atomic ratio determined by inductively coupled plasma atomic emission spectrometry (ICP-AES) analysis (Supplementary Fig. 10).

As a result of 2p core hole spin-orbit splitting, both experimental and theoretical Ni L-edge XAS spectra (Fig. 2c) are divided into two regions of interest: The L₃-edge near 856 eV and the L₂-edge near 874 eV. High-resolution sXAS spectra can also resolve the fine multiplet structure arising from Ni 2p-3d interactions and crystal field effects from the modulated local coordination environment^{34, 36}. We first measured Ni L-edges using partial fluorescent yield (PFY) on samples prepared *ex situ*, where we observed no difference compared to open-circuit potential conditions (Fig. 2d). We then acquired PFY at the Ni L-edge, with a probing depth of approximately 300 nm, collecting real-time spectra while applying constant anodic bias of +1.6 V, +1.8 V and +2.0 V vs. reversible hydrogen electrode (RHE), respectively.

In order to understand the changes in the Ni L-edge, we carried out first-principles calculations, solving the Bethe-Salpeter equation (BSE) for correlated electrons and holes with the goal of predicting theoretical sXAS spectra with the aid of the OCEAN simulation package^{38, 39}. The BSE is capable of reproducing the branching ratio of the L₃/L₂ edge correctly³⁸, together with the edge-shift in different chemical environments or oxidation states⁴⁰. We use ground-state DFT+U Kohn-Sham orbitals as input for the BSE calculation, with U=6.45 eV, identical with our formation energy calculations above. Our BSE calculations, working within a basis of single electron-hole pair excitations, cannot describe atomic multiplet splitting. In particular, for the Ni²⁺ system, the L₃-edge is split into a main peak and a secondary feature at higher energies. To reproduce multiplet effects, we further perform an atomic configuration

calculation with the CTM4XAS code^{41,42}. The magnitude of the crystal field splitting is chosen to be 2.0 eV, which is estimated from the DFT+U calculation for the layered Ni(OH)₂ structure. With the multiplet effects included, the simulated spectrum of Ni²⁺ is in excellent agreement with experiments.

These *in situ* sXAS studies provide the first direct view of Ni⁴⁺ as it forms during the operation of oxyhydroxide catalysts during water oxidation. We performed NiO (Ni²⁺), LiNiO₂ (Ni³⁺), and potassium nickel(IV) paraperiodat K₂Ni(H₂IO₆)₂ (Ni⁴⁺) references at Beijing Light Source 4W1B beamline (Supplementary Fig. 4). We chose LiNiO₂ as our Ni³⁺ reference because it has a layer double hydroxide structure. We subtracted Ni²⁺ from potassium nickel(IV) paraperiodat K₂Ni(H₂IO₆)₂ (Ni⁴⁺) reference and obtained single-peak shape similar to our simulated spectra. The intensity ratio of the double-peak features in the Ni-L3 region, at 855.6 and 857.5 eV, provide one qualitative proxy for the relative proportion of the two Ni oxidation state populations of interest, i.e. they capture the progression from Ni²⁺ to Ni³⁺ and Ni⁴⁺ upon application of a potential in the NiCoFeP sample (Supplementary Fig. 12). The emergence of an additional, broader peak at the Ni L2-edge around 875.1 eV confirms the generation of Ni⁴⁺ species (Fig. 2d). The relative L3-edge chemical shift between Ni²⁺ and Ni⁴⁺ are also consistent with our BSE calculation, which predicts a blue shift of 2.6 eV from Ni²⁺ to Ni⁴⁺ (dashed curves in Fig. 2c). Both the Ni L3- and the L2-edge sXAS revert back to their original state when a cathodic bias (0 V vs. RHE) is applied in the NiCoFeP sample. This suggests a return to the low Ni oxidation state and paints a picture of the reversible redox of Ni⁴⁺/Ni²⁺ (Fig. 2d).

We also acquired Ni *in situ* sXAS while varying potentials for NiCoP (Supplementary Fig. 13) and NiP (Supplementary Fig. 14). The NiCoP sample exhibited a significantly lower $\text{Ni}^{4+}/\text{Ni}^{2+}$ ratio than the NiCoFeP sample at varying potentials (Fig. 2e and Supplementary Fig. 13). In NiP, we detect no ostensible changes in the oxidation state even in presence of a 2.0 V potential (Supplementary Fig. 14), consistent with the DFT prediction of the high formation energy of Ni^{4+} in NiP.

Overall, the *in situ* studies agree with a physical picture in which Ni is the ionic species most susceptible to become oxidized to high valence in NiCoFeP; and that the incorporation of Co, Fe and P beneficially modulates the electronic structure of Ni metal oxides. It results in a reduced formation energy for Ni^{4+} , a configuration favorable for water oxidation, consistent with the DFT studies in this work.

We also used *in situ* XAS at the Fe and Co L-edges of NiCoFeP and relevant controls. *In situ* spectra of Fe L-edge indicate that Fe^{3+} remains the dominant valence state before and during electrochemical operation (Supplementary Fig. 15), while the valence state of Co evolves from Co^{2+} under open circuit to Co^{3+} upon application of a potential, including only a modest admixture with Co^{2+} in all samples (Supplementary Fig. 16).

Water oxidation electrocatalysis. We then proceeded to compare the catalytic performance of NiCoFeP with that of the reference samples NiP, NiCoP and IrO_2 for the oxygen evolution reaction. To increase the conductivity of the electrode, we loaded our catalysts on a gold-plated nickel foam. Representative OER currents were measured in CO_2 -saturated 0.5 M KHCO_3 aqueous electrolyte at a scan rate of 1 mV s^{-1} (Fig. 3a)

(currents are uncorrected and thus are already penalized by the effects of resistive losses incurred within the electrolyte).

The NiCoFeP-on-gold-plated Ni foam requires an overpotential of only 330 mV at 10 mA cm⁻² (Table 1, all current densities based on projected geometric area), which is 70 mV lower than that of the best commercial precious metal oxide IrO₂ catalyst tested under identical conditions. To the best of our knowledge, this is the first report of an earth-abundant catalyst that consistently outperforms precious metal oxide catalysts in a pH-neutral medium^{3, 5-7} (Table 1). To evaluate the effect of surface area using an additional independent method, we analyzed the surface area via the Brunauer Emmett and Teller (BET) technique. This permitted us to report normalized kinetic current density (referred to as specific activity) as a function of potential versus RHE in order to assess the intrinsic activities of the catalysts. Using this metric, the intrinsic activity of NiCoFeP oxyhydroxide is confirmed to be notably higher than that of the controls (Supplementary Fig. 17). From BET, the high catalytic activity of NiCoFeP oxyhydroxide includes an appreciable intrinsic (not surface-area-related) contribution.

To assess the kinetic barriers involved in OER, we studied the effect of temperature on the performance of the catalysts (Fig. 3b and Supplementary Fig. 19). OER proceeds more rapidly at elevated temperatures, with an exponential temperature dependence of the chemical rate constant⁴³. The electrochemical activation energies extracted from the slope of the Arrhenius plots are lower for the new catalyst than that of the precious metal oxide IrO₂ (Table 1 and Supplementary Fig. 16). The most active NiCoFeP catalysts in this study exhibit the apparent barrier value of 27 kJ mol⁻¹, indicating the high catalytic activity.

Our studies suggest that developing a TM oxyhydroxide having lower Ni^{4+} formation energy provides an effective strategy and new guideline to enhance OER catalytic activity (Fig. 3c). To characterize performance stability, we operated the OER catalyst on gold-plated Ni foam under constant current density of 10 mA cm^{-2} continuously for 100 hours. We observed no appreciable increase in potential (Fig. 3d) over this time interval. To verify that the catalyst remained physically intact, we checked using ICP-AES for any observable leaching of the constituents into the electrolyte. We were able to detect no measurable elemental loss using this sensitive method. Consistently high activity throughout the entire duration of stability testing was further confirmed by measuring the rate of O_2 evolution from the NiCoFeP-on-gold-plated Ni foam catalyst. We obtained quantitative (i.e. unity Faradaic efficiency) gas evolution of oxygen to within our available $\pm 5\%$ experimental error. Our approach to minimize the formation energy required for active metal sites may also promote further avenues to other related oxidation reactions.

We then assembled an electrolyzer that converts CO_2 and H_2O into CO and O_2 based on NiCoFeP oxyhydroxides as OER catalysts and recently-reported Au nanoneedles as CO_2 reduction catalysts⁴⁴. We achieved a 1.99 V cell voltage at 10 mA/cm^2 , with an electricity-to-chemical-fuel efficiency of 64% (Figure 3e). This was accomplished in a membrane-free system that avoids membrane resistance. In contrast, when we use an H-cell electrolyzer with bipolar membranes, which enabled the use of state-of-art OER catalyst FeCoW in a favorable alkaline electrolyte, the total cell potential reached 2.37 V and achieved a 54% electricity-to-chemical-fuel efficiency (see Supplementary Table 3). The new neutral OER catalysts reported herein provide

efficient membrane-free electrolysis, and thus represent an overall enabling component in CO₂ reduction.

Methods

Energetics Calculations. Bulk energetics of metal and phosphorous doped Ni-oxides were calculated using density functional theory with the Hubbard-U framework (DFT+U) to account for the strongly localized d-electrons for Ni, Fe, and Co. The Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional and Blöchl's all-electron, frozen-core, projector augmented (PAW) method with a plane-wave basis set cutoff value of 750 eV were used with the Vienna Ab-initio Simulation Package (VASP). The Gibbs free energy of formation, ΔG°_f , was calculated using the framework of Persson et.al¹⁶ where the reference energies are taken from the most stable phase of each element. Full details are given in Supplementary Information.

Synthesis of gelled NiCoFeP oxyhydroxides and controls. All gelled oxyhydroxides were synthesized using a modified aqueous sol-gel technique¹⁶.

In-situ X-ray absorption. X-ray absorption measurements at the nickel, Co and iron L-edges were performed at the spherical grating monochromator (SGM) beamline 11ID-1 at the Canadian Light Source. The window of the sample cells was mounted at an angle of roughly 45° with respect to both the incident beam and the detectors. The bodies of the sample cells were fabricated on an Object Connex500 printer by 3D printing with DurusWhite material. The catalysts were coated on the windows via drop-casting with a thickness of 400 nm. The gold-covered window was contacted by gold wires. Silver and platinum wires were used as reference and counter electrodes. All measurements were made at room temperature in the fluorescence mode using Amptek silicon drift

detectors (SDDs) with 1024 emission channels (energy resolution ~ 120 eV). Four SDDs were employed simultaneously. The scanning energy ranges of Ni, Co and Fe L-edges were set between 840 and 900 eV, 760 and 820 eV, and 690 and 750 eV, respectively. The partial fluorescence yield (PFY) was extracted from all SDDs by summation of the corresponding metal L emission lines. Full details can be found in Supplementary Information.

L-edge simulation of XAS with the OCEAN code. The formalism of the Bethe-Salpeter equation can be found in Ref (38). The input DFT+U calculations are performed using norm-conserving pseudopotentials. Very large energy cutoff of 160 Ryd is used for the plane-wave basis set of the Kohn-Sham wavefunctions. The key part of the electron-core-hole interactions is the screened Coulomb interactions, which is obtained within the random phase approximation. A large number of empty bands up to 50 eV above the Fermi level are included in the calculations. A 6x6x6 k-grid over the Brillouin zone is used to produce the sXAS spectra, which is convoluted with Lorentzians of 0.5 eV.

Electrochemical Measurements. Electrochemical measurements were performed using a three-electrode system connected to an electrochemical workstation (Autolab PGSTAT302N). Cyclic voltammetry (CV) measurements at 50 mV s^{-1} were performed for 3 cycles prior to the record of linear scan voltammetry (LSV) at 1 mV s^{-1} for each sample. The LSV performance of NiCoFeP, NiCoP, NiP and the corresponding controls with no phosphorus were also normalized by the BET surface area. The apparent electrochemical activation energy (E_a) for water oxidation was determined using the Arrhenius relationship. The calibrations of Ag/AgCl reference electrode and Ag wire

reference electrode were conducted in the standard three-electrode system (the same system as that for performance measurements) as reference electrodes, using Pt foil as working and counter electrodes. Full details are given in Supplementary Information.

Data availability. The authors declare that the data supporting the findings of this study are available within the article and its Supplementary Information files, or from the corresponding author on reasonable requests. Detailed Supplementary Information is available online containing descriptions of methods; Supplementary Figs. 1–29 with results and discussions; assessment of electronic structures; DFT calculations.

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Author contributions

E.H.S., D.P. and B.Z. supervised the project. X.Z. and B.Z. designed and carried out the experiments. X. Z., B. Z. and R. C. performed the soft X-ray measurements. P. D. L., Y. L., D.P. and O. V. carried out simulation parts. All authors discussed the results and assisted during manuscript preparation.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permission information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to E. H. S. and B. Z.

Completing financial interests

The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper.

Fig. 1. Change in Gibbs Free Energies for $\text{Ni}^{2+} \rightarrow \text{Ni}^{3+} \rightarrow \text{Ni}^{4+}$. DFT calculated free energies for the oxidation of $\text{Ni}(\text{OH})_2$ to NiO_2 (right-hand-side column) with various Ni-site substitutions (if any): Ni (red), NiCo (green), NiCoFe (purple). Phosphorus and metal doping promotes the formation of NiO_2 (Ni^{4+}) by lowering the ΔG required for this transition. Doping with phosphorus lowers the change in formation energy for the $\text{Ni}(\text{OH})_2 \rightarrow \text{NiOOH}$ transition (left shift). DFT reveals that the phosphorus dopant prefers to be four-fold coordinated and introduces steric strain in the $\text{Ni}(\text{OH})_2$ structure as shown by the structure zoom-in of P (purple) and Ni (blue) centers in a P-doped $\text{Ni}(\text{OH})_2$ bulk crystal. Similarly, doping with Fe and Co lowers the change in formation energy for the $\text{NiOOH} \rightarrow \text{NiO}_2$ transition (down shift).

Fig. 2. The *in situ* electrochemical flow cell enabling *in situ* sXAS. (a, b) Schematic illustration of the *in situ* liquid cell experimental setup and the STEM image of NiCoFeP catalyst. A Si_3N_4 membrane (~ 100 nm) separates the liquid from the high vacuum region where soft x-rays propagate. Detection of x-ray absorption was performed by collecting x-ray fluorescence on the vacuum side. (c) *In situ* experimental (solid) and theoretical (dash) sXAS at the Ni L-edge of Ni^{2+} (brown) and Ni^{4+} (green), respectively. The theoretical sXAS is aligned according to L3 edge of Ni^{2+} . The top spectra obtained for NiCoFeP at +1.8 V vs. RHE during reaction was fitted by experimental sXAS of NiO (Ni^{2+}), LiNiO_2 (Ni^{3+}), and potassium nickel(IV) paraperiodat $\text{K}_2\text{Ni}(\text{H}_2\text{IO}_6)_2$ (Ni^{4+}) references after subtracting Ni^{2+} signals. (d) Ni L-edge of NiCoFeP and relevant controls. (e) $\text{Ni}^{4+}/\text{Ni}^{2+}$ ratio versus potential for NiCoFeP and controls.

Fig. 3. Performance of NiCoFeP oxyhydroxides catalysts and controls in a three-electrode configuration in CO₂ saturated 0.5 M KHCO₃ aqueous electrolyte. (a), The OER polarization curve of catalysts loaded on Au foam at 1 mV/s scan rate. (b), Arrhenius plot of the inverse temperature versus the log of the exchange current for NiCoFeP oxyhydroxides and controls on Au (111). (c), The relation between overpotential at 10 mA/cm² (left), activation energy (right) and the theoretical formation energy of Ni²⁺ to Ni⁴⁺ in NiCoFeP and controls. The solid lines are shown for guidance only. (d), Chronopotentiometric curves obtained with the NiCoFeP on gold-plated Ni foam electrode with constant current densities of 10 mA cm⁻², and the corresponding Faradaic efficiency from gas chromatography measurement of evolved O₂ (Each point was measured five times independently to obtain the average value and the absolute error). (e), Cell potential as function of time for CO₂ electrolysis at controlled current density 10 mA/cm² (Supplementary Fig. 22 (a) scheme of NiCoFeP-Au needles electrolysis (black line); (b) scheme of FeCoW-Au needles with bipolar membrane (red line)).

Table 1. Comparison of catalytic parameters of NiCoFeP and controls.

Samples	pH	On gold foam	On glass carbon	On Au(111)	Reference
		Overpotential (mV) at 10 mA/cm ²	Overpotential (mV) at 1 mA/cm ²	ΔH (kJ mol ⁻¹) at $\eta=700$ mV	
NiP	7.2	588	390	54.14	This work
NiCoP	7.2	547	330	37.36	This work
NiCoFeP	7.2	330	276	27.00	This work
Commercial IrO ₂	7.2	400	-	29.41	This work
IrO ₂	7.2	460 ^a	290 ^a	--	Ref (3)
MnO ₂	7.5	--	417 ^b	--	Ref (43)
Ni-based	6.8	--	467	--	Ref (7)
Co-Pi	7.0	--	550 ^b	--	Ref (6)

a: obtained the current density on Ti plate;

b: obtained the current density on a conductive glass substrate.